

Assignment No: - No. 7

Design a web application for e-commerce or students feedback review systems or any other applications with MongoDB Backend. using React.js and Express.

Name: - Piyush Rajendra Waghmare

PRN: - 123B1B284

DIV: - D4

Objectives:

- To develop a responsive, full-stack Student Feedback Review System tailored for institutional evaluation.
- To implement secure, role-based authentication (Student vs. Admin) using JSON Web Tokens (JWT) to protect user credentials and feedback integrity.
- To create an intuitive student dashboard that allows users to dynamically select their enrolled courses and respective faculty members.
- To design a seamless, interactive feedback form with star ratings and qualitative comments, providing immediate UI feedback (popups) upon successful submission.
- To manage and persist all system data (users, courses, faculties, and feedback records) securely using a NoSQL database (MongoDB).

Outcomes:

- **Data Persistence:** Successful storage and retrieval of student credentials, faculty assignments, and submitted feedback documents using MongoDB and Mongoose ODM.
- **Security & Authorization:** Implementation of robust JWT authentication to ensure that only verified students can submit feedback, and route guards to prevent unauthorized access to the dashboard.
- **Interactivity:** A dynamic React-based frontend that updates UI states (dropdown selections, popup notifications, rating components) efficiently without page refreshes.
- **Scalable API Design:** Development of a RESTful backend using Node.js and Express.js to handle login requests, fetch course/faculty mappings, and process feedback submissions.

Theory Concept in brief:

- **MERN Architecture:** A modern 3-tier web architecture consisting of a React.js frontend (Presentation Layer), Node/Express.js backend (Application Layer), and MongoDB (Data Layer).

- **RESTful APIs:** Stateless communication between the React frontend and Express backend using standard HTTP methods (POST for login/submitting feedback, GET for fetching faculty/courses).
- **JWT Authentication:** A stateless, token-based authentication mechanism where the server generates a token upon successful login, which the React client stores and sends in subsequent requests to access protected routes.
- **NoSQL Document Database:** Using MongoDB to store data in flexible, JSON-like documents, making it highly efficient for storing hierarchical or unstructured data like variable feedback form parameters.

Technology Used

Category	Technology / Tool	Purpose in Project
Frontend	React.js, HTML5, CSS3	Building a dynamic, responsive Single Page Application (SPA) for the student dashboard and feedback forms.
Backend	Node.js, Express.js	Handling RESTful API requests, business logic, and routing.
Database	MongoDB, Mongoose	NoSQL database for flexible and scalable storage of user data, courses, and feedback records.
Authentication	JSON Web Tokens (JWT), Bcrypt	Securing API endpoints, hashing passwords, and managing student/admin login sessions statelessly.
State Management	React Hooks (useState, useEffect)	Managing UI state, such as selected courses, dropdown menus, and popup visibility without page reloads.

API Client	Axios or Fetch API	Sending asynchronous HTTP requests from the React frontend to the Express backend.
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Dataset Used

Unlike machine learning projects that use static CSV datasets, this web application utilizes a dynamic, relational-like dataset stored in MongoDB in JSON/BSON format. The primary data entities include:

1. **Student/User Dataset:** Contains mock credentials (PRN, Email, Hashed Passwords) to authenticate users and prevent multiple submissions.
2. **Academic Mapping Dataset:** Contains Course Codes, Course Names, and Assigned Faculty members. This dataset populates the dynamic dropdowns on the student dashboard.
3. **Feedback Metrics Dataset:** The core operational dataset generated by users, containing quantitative ratings (1 to 5 stars) across various parameters (e.g., punctuality, subject knowledge) and qualitative text comments.

Functional Description

The Student Feedback Review System operates through a series of seamlessly integrated functional modules:

1. **Secure Authentication Module:** The application strictly restricts access. When a student attempts to log in, the Express backend verifies their PRN

and password against the hashed credentials in MongoDB. Upon success, a JWT is generated and sent to the frontend, which is then used to authorize access to the dashboard.

- 2. Dynamic Dashboard & Selection:** Once authenticated, the student is redirected to their dashboard. The React frontend makes a GET request to fetch the courses the student is enrolled in. The UI presents cascading dropdown menus where the student first selects a course, which dynamically filters and displays the respective faculty member assigned to that subject.
- 3. Interactive Feedback Submission:** After selecting the faculty, the interactive feedback form is rendered. It utilizes a visually appealing star-rating system (1 to 5) alongside a text area for constructive comments. When the student clicks "Submit", a POST request carrying the JWT and feedback data is sent to the backend.
- 4. Data Validation and Persistence:** The Express backend validates the incoming payload to ensure no empty fields or invalid ratings are submitted. It also checks the database to ensure the student hasn't already submitted feedback for this specific course. Once validated, the document is saved into the MongoDB Feedbacks collection.

5. Real-Time UI Feedback (Popups): Upon successful database insertion, the backend sends a 200 OK response. The React frontend immediately triggers a success popup (e.g., using Toastify or a custom modal) thanking the student, and resets the form, ensuring a smooth and responsive user experience.

Program Code with output Screenshot: -
Program code is attached with zip File :-

Student Feedback Review System.zip

Program Output: -

The screenshot displays a web application titled "Student Feedback Review" running on a browser at localhost:5173/student. The user is logged in as Piyush Waghmare (student). The interface includes a "Student Dashboard" with a welcome message and a "Feedback submitted successfully" notification. The main section is "Comprehensive Course and Faculty Evaluation", which allows users to select a course (currently "CC - Cloud Computing (Supriya Mahadevkar, Shreya Patil)") and evaluate it. The evaluation is divided into sections, with "Section A: Course Content & Structure" currently selected. Under this section, there are two evaluation items. The first item, "1. The course objectives and expected learning outcomes were clearly communicated at the beginning of the semester.", has a rating of 5. The second item, "2. The course content was logically organized and progressed at a comfortable pace.", is partially visible. On the right side, the "My Submitted Feedback" section shows a list of submitted feedback entries. The first entry is for "CC - Cloud Computing" with an overall course average rating of 3.7/5. The second entry is for "FSD - Full Stack Web Development" with an overall course average rating of 3/5. Each entry includes fields for "Positive aspects" and "Suggestions for improvement".

1. The course objectives and expected learning outcomes were clearly communicated at the beginning of the semester.

12345

2. The course content was logically organized and progressed at a comfortable pace.

12345

3. The provided study materials (textbooks, lecture slides, reference links, and handouts) were relevant, high-quality, and up-to-date.

12345

4. The course maintained a healthy balance between theoretical concepts and practical/hands-on applications.

12345

5. The overall workload for this course (assignments, reading, projects) was reasonable and manageable.

12345

Section B: Faculty & Teaching Effectiveness

Suggestions for improvement

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FSD - Full Stack Web Development

Instructor(s): .

Overall course average rating: 3/5

Positive aspects

No response provided.

Suggestions for improvement

No response provided.

BC - Blockchain Technology

Instructor(s): Unknown

Overall course average rating: 3.6/5

Positive aspects

It's a back way of teaching

Suggestions for improvement

There should be a faculty who will teach topic by topic everything

DAA - Design and Analytics of Algorithm

Instructor(s): Unknown

18. Overall, I feel confident that I have achieved the learning objectives of this course and can apply this knowledge in the future.

12345

Section E: Qualitative Feedback

19. What were the most positive aspects of this course and the instructor's teaching methodology?

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20. What specific suggestions do you have to improve the course content, assignments, or the instructor's teaching style?

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☐ Submit as anonymous

Update Feedback

localhost:5173/student

Student Feedback Review

localhost:5173 says

Are you sure you want to delete this feedback?

OKCancel

Student Dashboard

Welcome, Piyush Waghmare. Submit and manage your course feedback.

Feedback submitted successfully

Comprehensive Course and Faculty Evaluation

Your feedback is highly valued and will remain anonymous.

Select course

CC - Cloud Computing (Supriya Mahadevkar, Shreeya Patil)

Section A: Course Content & Structure

Evaluate syllabus, materials, and course design.

My Submitted Feedback

CC - Cloud Computing

Instructor(s): .

Overall course average rating: 3.7/5

Positive aspects

.....

18. Overall, I feel confident that I have achieved the learning objectives of this course and can apply this knowledge in the future.

1 2 3 4 5

Section E: Qualitative Feedback

19. What were the most positive aspects of this course and the instructor's teaching methodology?

Share positive aspects and highlights...

20. What specific suggestions do you have to improve the course content, assignments, or...

Share suggestions for improvement...

☐ Submit as anonymous

Update Feedback

Success!

Feedback submitted successfully.

OK

localhost:5173/login

School

Student Feedback Review

Login

Login

Login to access your dashboard.

Email

piyush.waghmare142@gmail.com

Password

.....

Login

[Don't have an account? Register](#)

Create Account

Register as a student to submit feedback.

Name

Piyush Waghmare

Email

piyush.waghmare142@gmail.com

Password

Register

[Already have an account? Login](#)

Cluster0

admin

local

student_feedback_review_system

courses

feedbacks

users

CLUSTERS (1)

Search clusters

Cluster0

admin

local

student_feedback_review_system

courses

feedbacks

users

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

[Options](#)

ADD DATA UPDATE DELETE EXPORT CODE

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1-6 of 6

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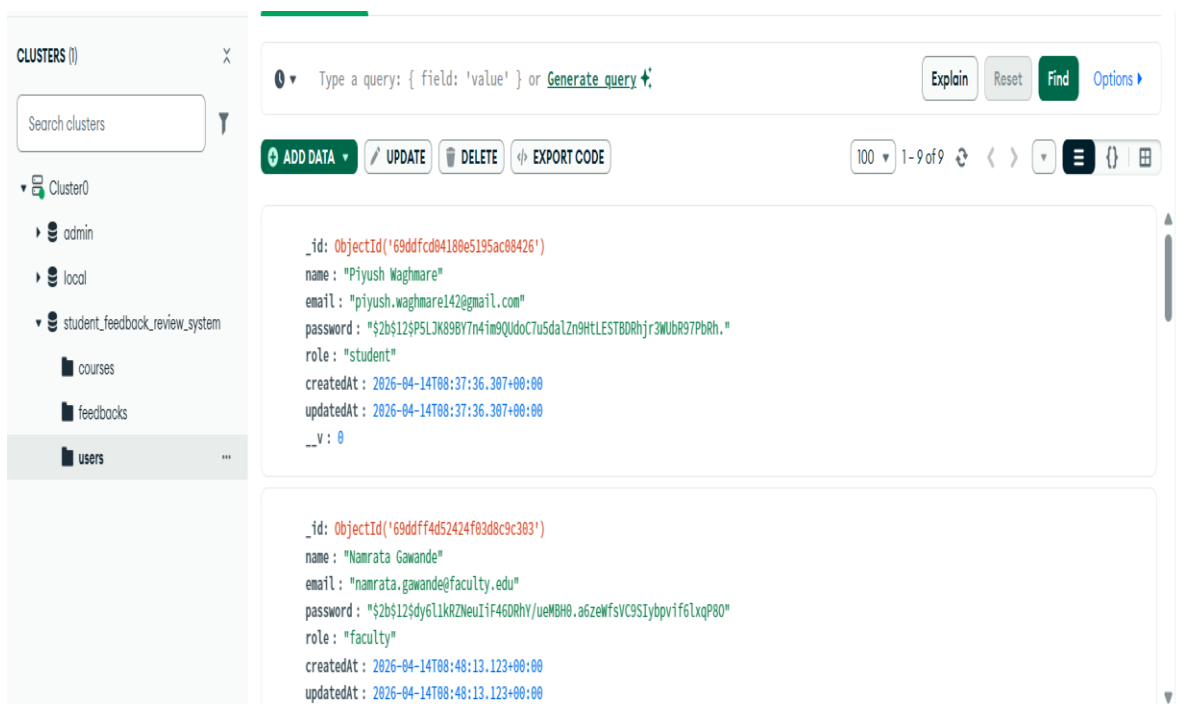
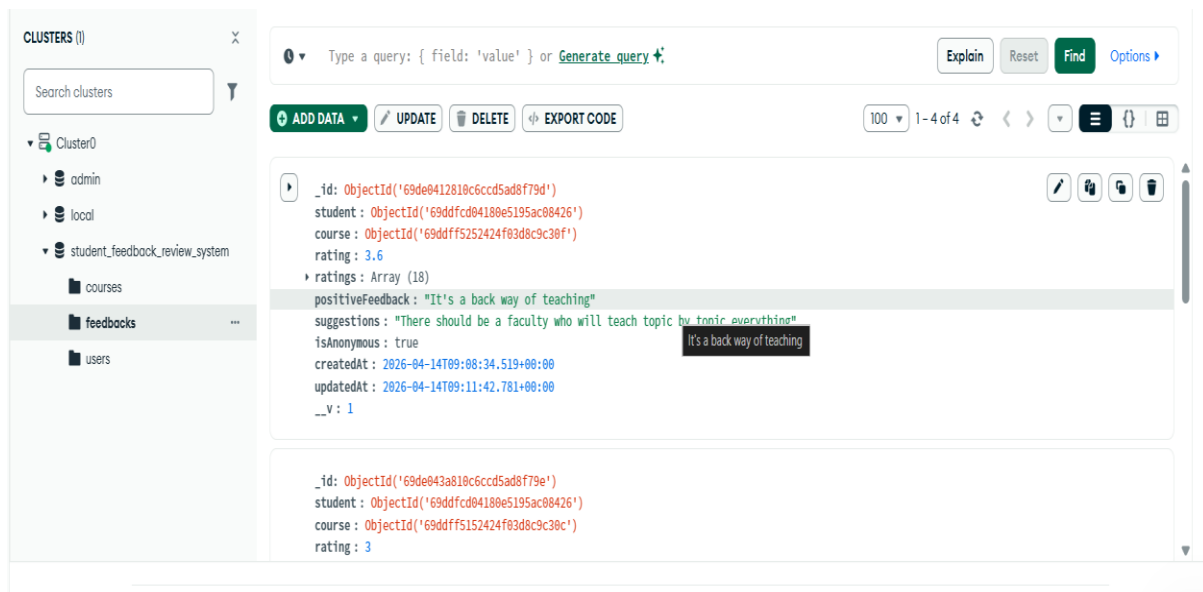
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description : ""
instructors : Array (1)
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  updatedAt : 2026-04-14T08:48:17.756+00:00
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  title : "Software Engineering"
  description : ""
  instructors : Array (1)
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    updatedAt : 2026-04-14T08:48:17.852+00:00
    __v : 0
```

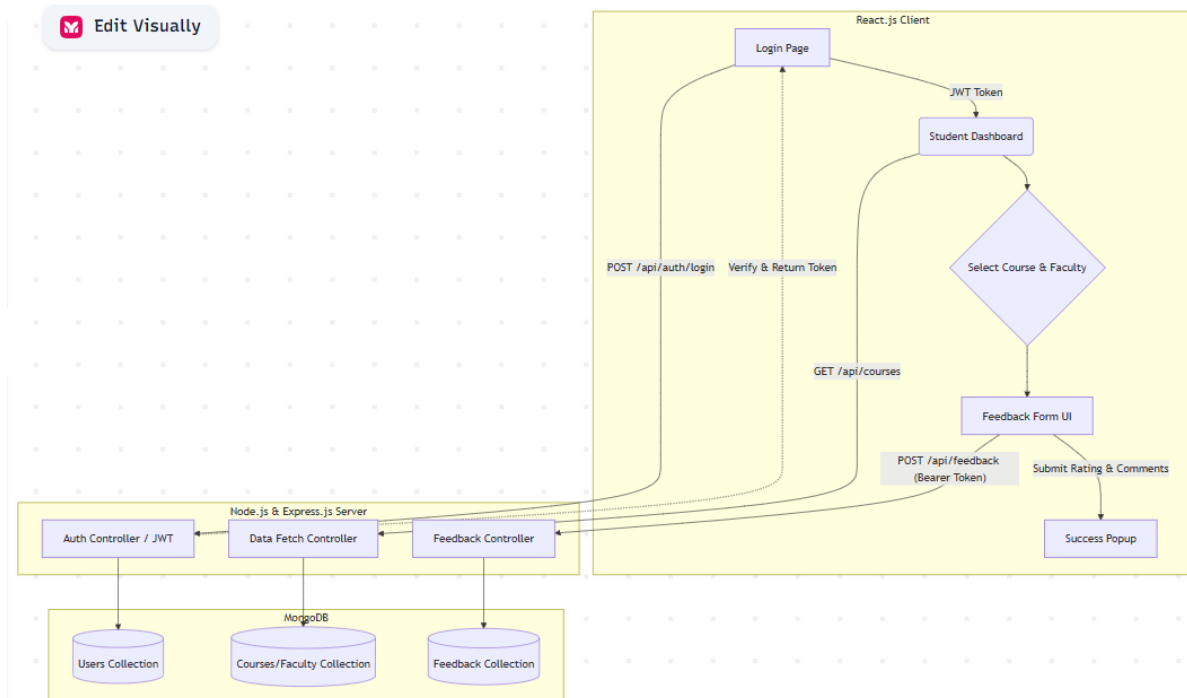
System Status: All Good

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System Architecture Diagram

Below is the Mermaid architecture code representing the data flow and system structure of the Student Feedback Review System:



Conclusion

The development of the Student Feedback Review System successfully demonstrates the power and efficiency of the MERN (MongoDB, Express.js, React.js, Node.js) stack. By utilizing React.js, the project achieved a highly interactive and seamless user interface that operates without jarring page reloads. The integration of Node.js and Express.js provided a robust backend architecture capable of handling RESTful API requests securely.

Furthermore, the implementation of JSON Web Tokens (JWT) ensured that the feedback process remained secure, anonymous, and restricted to authorized students only. The choice of MongoDB as the database allowed for highly flexible data modeling, easily accommodating the varying structures of user credentials and feedback forms.